**Biochemistry Cume** **October 10, 2008**

Molecular Biology

1. (20 points) Define each of the following terms.
   A. Restriction enzyme.
   B. Polymerase chain reaction.
   C. Topoisomerase.
   D. Frameshift mutation.

2. Two-milligram samples of DNA and RNA are incubated overnight in 0.05 M buffer at pH 9.9. Control DNA and RNA samples (also 2 mg) are incubated overnight in a different buffer (0.05 M) at pH 4.4. The next morning, aliquots of each sample are subjected to electrophoresis appropriate to separate nucleic acid molecules by size, and results are visualized by staining.

   A. (5 points) What would be an appropriate stain?

   B. (35 points) Explain qualitatively what results you expect to see from the four samples. Briefly account for expected sample-to-sample differences, if any.

3. Five-microgram samples of total *E. coli* chromosomal DNA and total human chromosomal DNA are each digested to an average size of 8,000 nucleotide pairs. Both samples are free of proteins. Each DNA sample is then heated (in aqueous solution at neutral pH) to 98° C for 20 minutes and then very slowly cooled to 38° C over a period of 5 days. During the cooling process, the absorbance at 260 nm is recorded continuously for each sample.

   A. (15 points) Why is a wavelength of 260 nm chosen?

   B. (25 points) Explain qualitatively what results you expect to see from the *E. coli* and human DNA samples. Briefly account for expected sample-to-sample differences, if any.